Preparation of the report.

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(English version by Sławomir Lasota)

Protocol Contents

- Protocol is an OFFICIAL document, not a draft.
- Exercise name.
- Names of persons performing the exercise.
- Group/time slot (e.g., Gr. 1 Tuesday 12:00).
- Date of execution.
- Measurement schematics or references to the exercise program
- Clearly and unambiguously recorded measurement results:

  table with description: measured quantity, unit
- Instructor's signature

Report Contents

- Title page.
  - Name of the exercise performed.
  - Names and surnames of persons performing the exercise.
  - Distinguished author – the responsible person who will receive credit or report return.
  - Group / execution term.
  - Execution date.

- Exercise objective – briefly what was performed and why (without theoretical introduction!).
- Numbered subsections corresponding to performed exercise program points containing proper processing of measurement results.
- Conclusions.

Data Processing Structure

- Subsection number and title.
- Brief description of what was done in the exercise vs what was done in the report.
- Measurement schematic with marked measured quantities.
- If the protocol is too illegible, rewritten measurement results from the protocol.
- Plotted characteristics based on measurements.
- Calculations, e.g., uncertainties or parameters:
  - Formula according to which calculations were performed.
  - Example calculations with substitutions into the formula of data from the protocol.
- Table with calculations.

Sample report

The objective of the exercise was to investigate the digital multimeter type xxx used in the laboratory for measuring DC voltages and currents as well as resistance.

1. Determining relative and absolute voltage measurement uncertainty

The measurement setup diagram is shown in Fig. 1.1.

Figure

Fig. 1.1. Measurement setup diagram for determining voltage measurement uncertainty

Measurements were conducted for three measurement ranges: $200mV$, $2V$, $20V$. In each range, 10 measurements were performed from the maximum value close to the measurement range to the minimum value constituting $0.05 \ldots 0.01$ of the measurement range. Measurement results were recorded in the Exercise protocol and copied to Table 1.1.

Exercise Objective

Subsection Number and Title

Measurement Schematic

Description of What Was Done During the Exercise

Sample Report (continued)

Determined for individual ranges δ_m and $(|\Delta_R| + |\Delta_q|)$ were placed in Table 1.1.

For calculations of absolute measurement uncertainty Δ and relative measurement uncertainty δ, formulas (1.1) and (1.2) were used respectively:

$$\Delta = \pm(|\delta_m \cdot U_m| + |\Delta_R| + |\Delta_q|) \quad (1.1)$$

$$\delta = \pm(\Delta / U_m) \cdot 100\% \quad (1.2)$$

The formula according to which calculations were made

Obtained for each measurement point results were recorded in appropriate columns of Table 1.1.

Table with calculations

Table 1.1. Summary of measurement and calculation results

range	I_{EFM}	multiplier	U_m [...]	δ_m [%]	$\|\Delta_R\| + \|\Delta_q\|$	Δ [...]	δ [%]
$200\ mV$	... ...			$0.5\ \%$	$\pm 0.1\ mV$		

Example substitutions into the formula

$$\Delta = 0.5\% \cdot 180mV + 0.1mV = 0.9mV + 0.1mV = 1mV$$

Plot

- Full name unambiguously describing the plotted relationship.
- Axes:
  - labeled: measured quantity, unit
  - allowed scales: linear (with "round" step) or logarithmic.
- Shaded/dashed grid in background enabling readout of any point.
- All measurement points marked (consistent with protocol).
- Characteristic approximation based on measurement points (do not include gross errors).
- Legend – if plot contains more than one curve (characteristic, waveform, etc.).

Types of Plots

- A plot is a mathematical function
- In technology and science, we distinguish certain special functions:
  - Time waveform is the change of some quantity (e.g., voltage) over time, i.e., the X axis is the time axis (t).
  - Static characteristic is the dependence of one quantity on another (e.g., voltage vs current: current-voltage characteristic, input characteristic, output characteristic, etc.)
  - Dynamic characteristic is the dependence of some quantity on frequency (or angular frequency), e.g., amplitude-frequency characteristic (amplitude spectrum).

A characteristic is a feature of the system/element, whereas time waveforms may differ from each other.

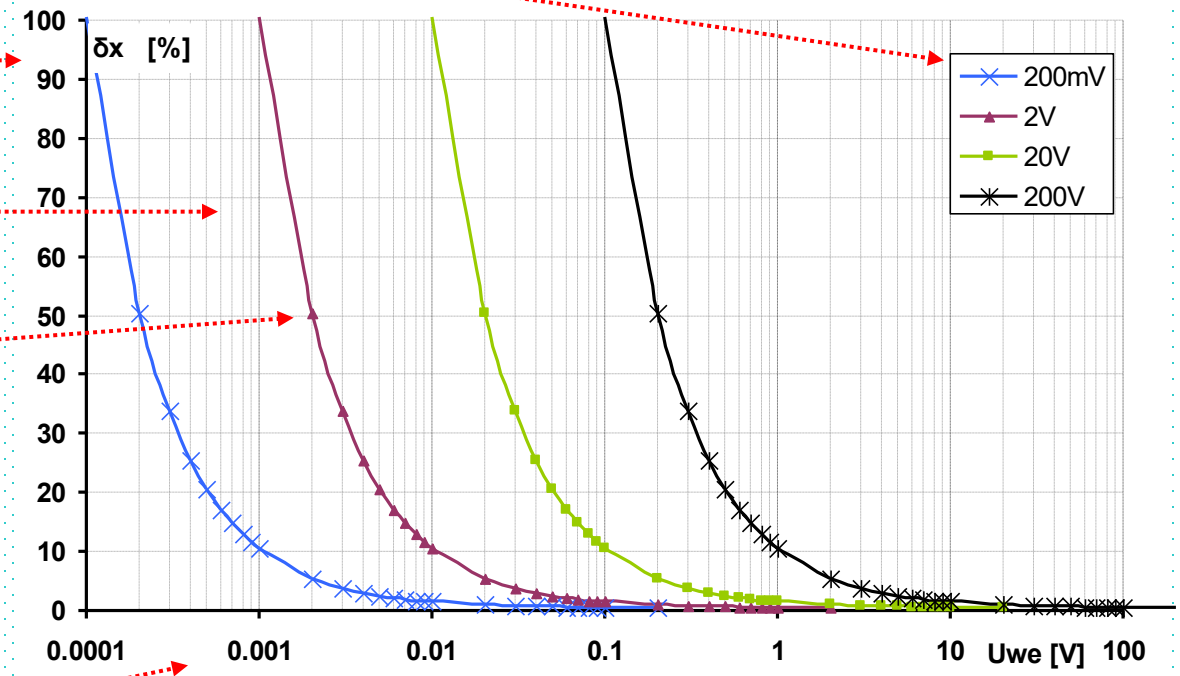
Example of a correct plot

Legend

Axis in linear scale

Grid

Measurement points



Axis in logarithmic scale

Fig. 1. Relative uncertainty vs input voltage.

Caption

Example of incorrect plot (1)

Lack of axis description (measured quantity, unit)

Lack of vertical grid

Unnecessary background, overly prominent horizontal grid

Axis with labels instead of values

(Linear chart type, not XY)

Niepełna nazwa

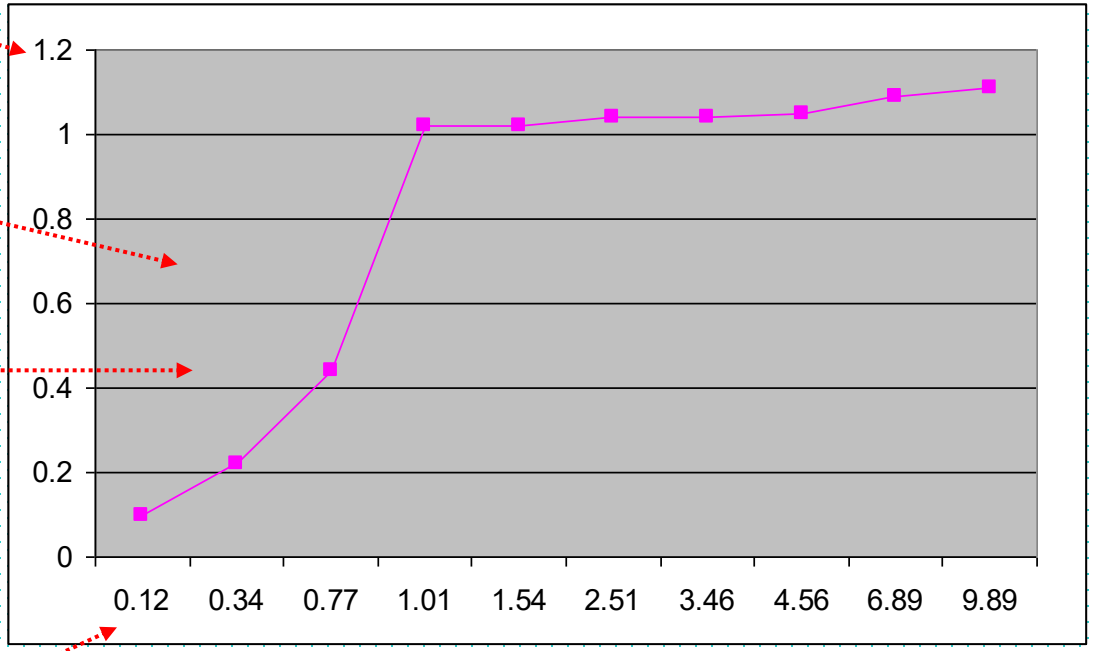


Fig. 1. Output characteristic

If names do not refer to quantities marked on the schematic, they are too unspecific (should be e.g., U_{CE} and I_C).

Lack of units

U_1	I_I
0.12	0.1
0.34	0.22
0.77	0.44
1.01	1.02
1.54	1.02
2.51	1.04
3.46	1.04
4.56	1.05
6.89	1.09
9.89	1.11

Example of incorrect plot (2)

Lack of parameter at the curve or legend

Very poorly chosen scale

Somewhat poorly chosen scale (sometimes acceptable)

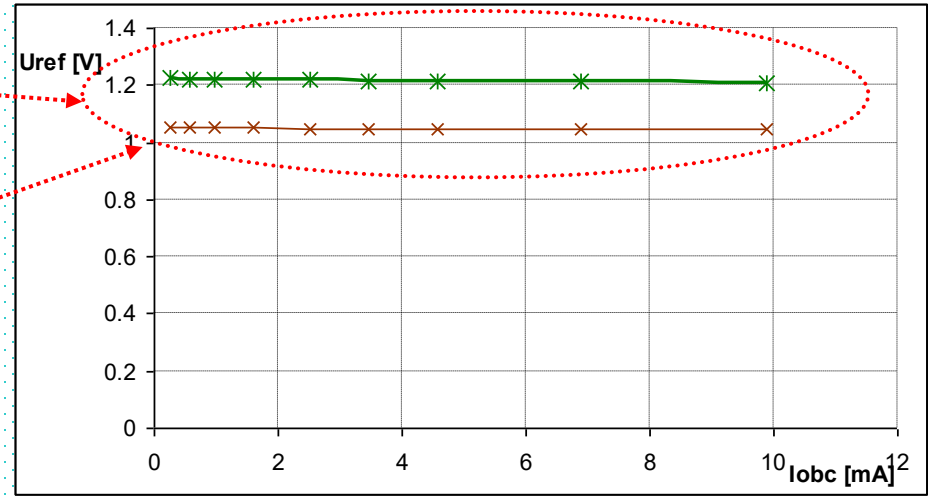
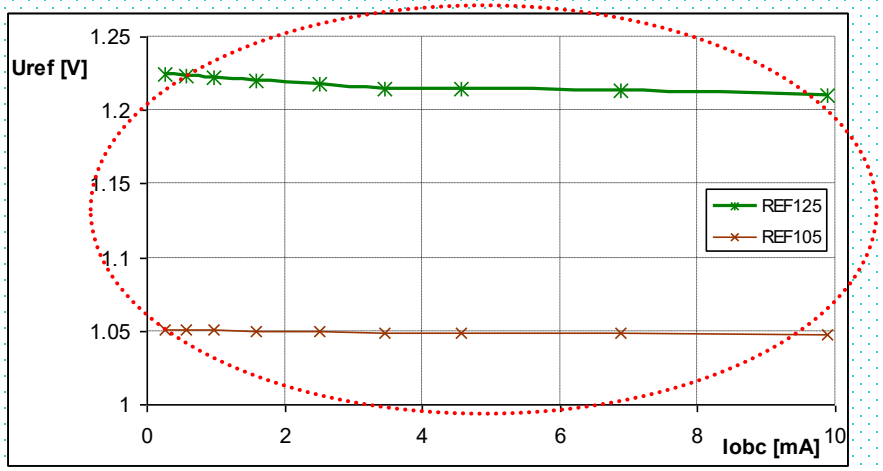


Fig. 1. Output characteristics of voltage reference sources.



Iobc [mA]	Uref1[V]	Uref2[V]
0.26	1.051	1.225
0.57	1.0505	1.223
0.96	1.0504	1.222
1.59	1.0503	1.22
2.51	1.05	1.218
3.46	1.0492	1.215
4.56	1.0489	1.2145
6.89	1.0485	1.213
9.89	1.0481	1.21

Example of how to correct the plot

Dual scale with unambiguously assigned curves:

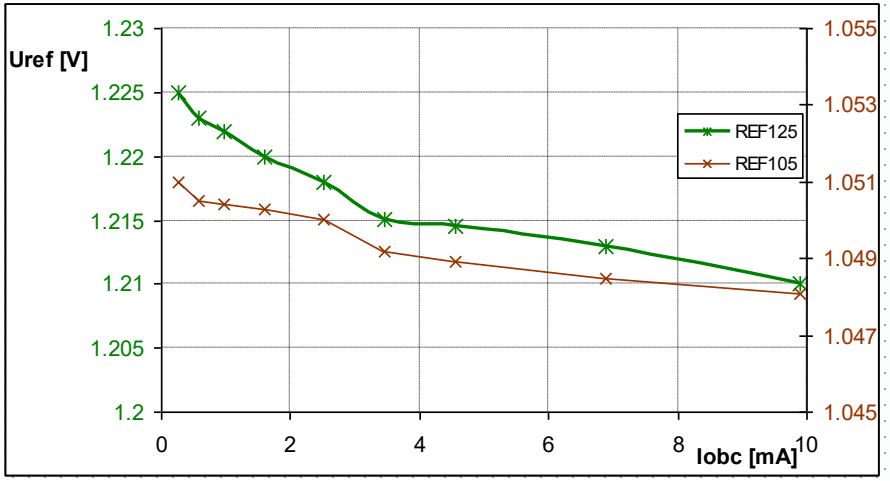


Fig. 1. Output Characteristics of Reference Voltage Sources.

Or simply 2 drawings:

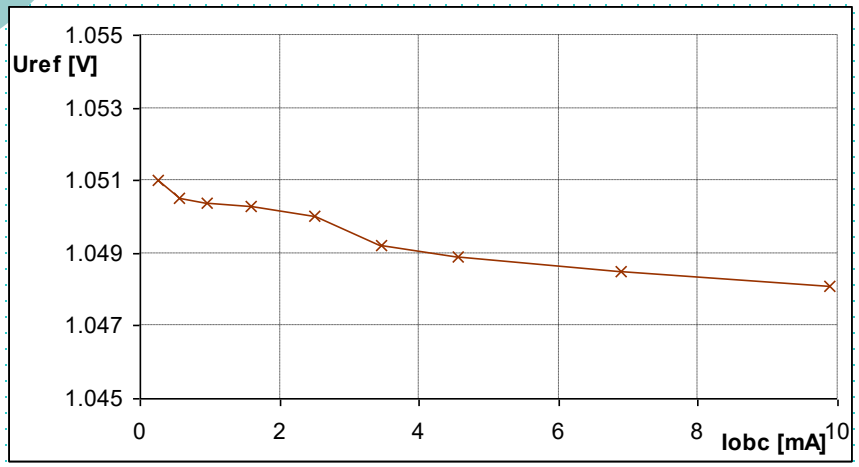


Fig. 1. Output Characteristic of REF105 Reference Voltage Source.

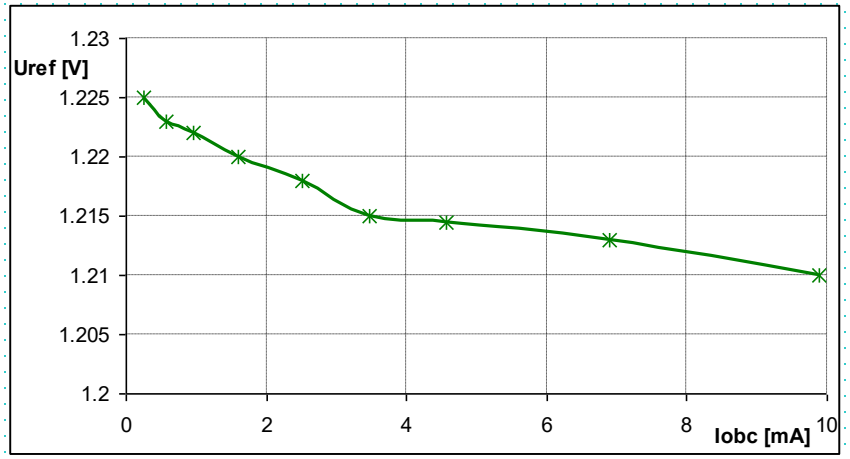


Fig. 2. Output Characteristic of REF125 Reference Voltage Source.

Example of incorrect plot (3)

Default curve name in unnecessary legend

Poorly chosen unit (should be mA)

Poor approximation

(according to the table, the function is monotonic)

Lack of measurement points

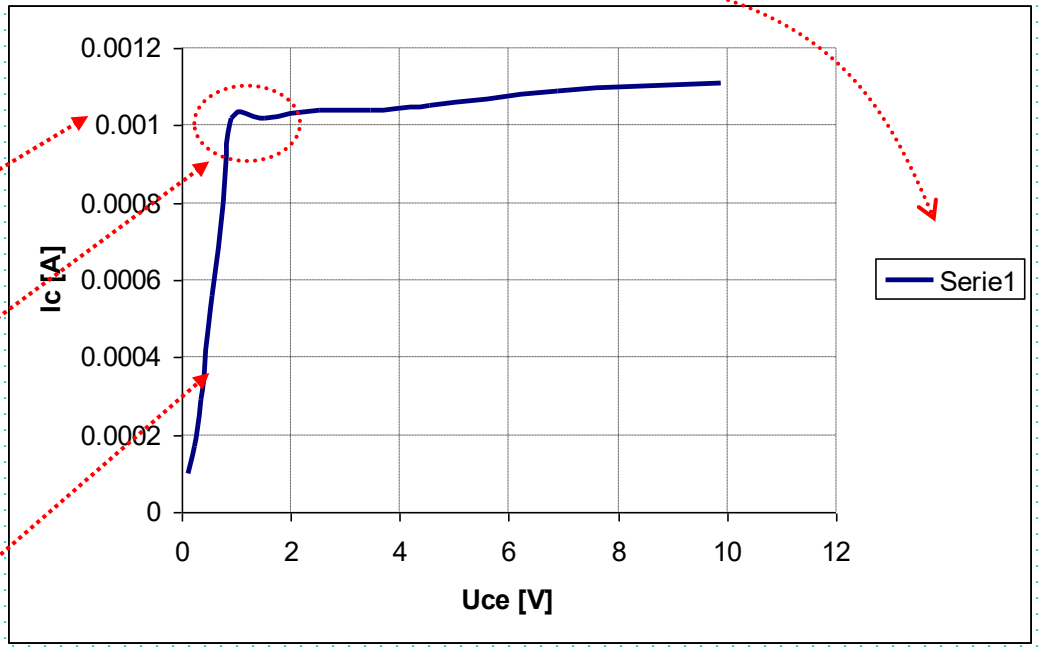


Fig. 1. Output characteristic of npn transistor BC107

Uce [V]	Ic[A]
0.12	0.0001
0.23	0.00017
0.34	0.00029
0.56	0.00056
0.77	0.0008
0.91	0.00102
1.54	0.00102
2.51	0.00104
3.46	0.00104
4.56	0.00105
6.89	0.00109
9.89	0.00111

Example of how to correct the plot

Uce [V]	Ic[A]	Ic[mA]
0.12	0.0001	0.1
0.23	0.00017	0.17
0.34	0.00029	0.29
0.56	0.00056	0.56
0.77	0.0008	0.8
0.91	0.00102	1.02
1.54	0.00102	1.02
2.51	0.00104	1.04
3.46	0.00104	1.04
4.56	0.00105	1.05
6.89	0.00109	1.09
9.89	0.00111	1.11

Auxiliary unit conversion

Unit selected to minimize the number of zeros

Manual approximation that captures the shape while preserving table features

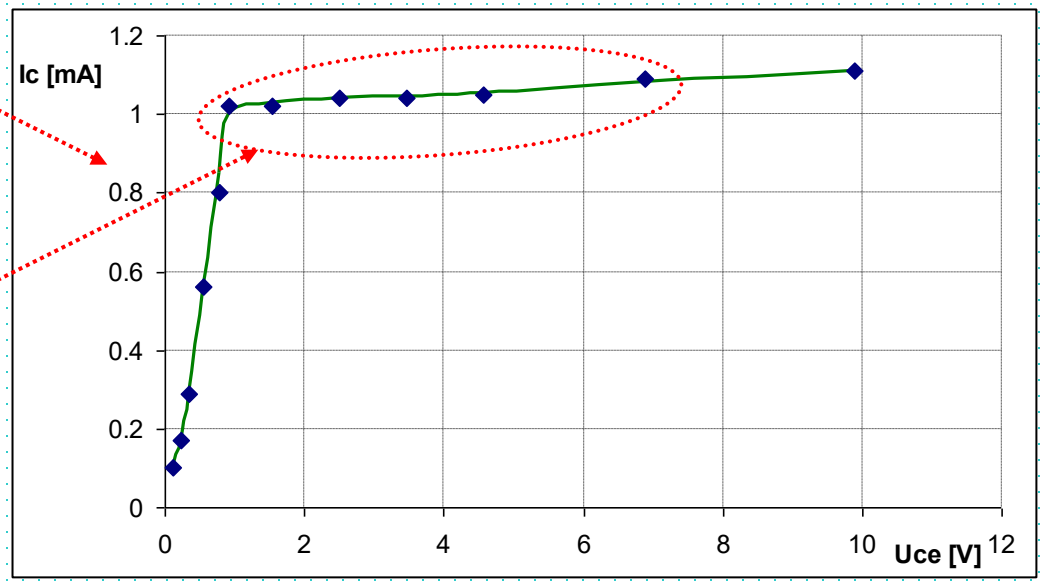


Fig. 1. Output characteristic of npn transistor BC107

Example of how to handle a gross error

Uce [V]	Ic[mA]
0.12	0.1
0.23	0.17
0.34	0.29
0.56	0.56
0.77	0.8
0.91	1.02
1.54	1.02
2.51	1.04
3.46	1.4
4.56	1.05
6.89	1.09
9.89	1.11

Gross error

Gross error:

1. Mark in the table.
2. Mark on the plot, exclude from approximation.
3. Mention it in conclusions and try to explain it.

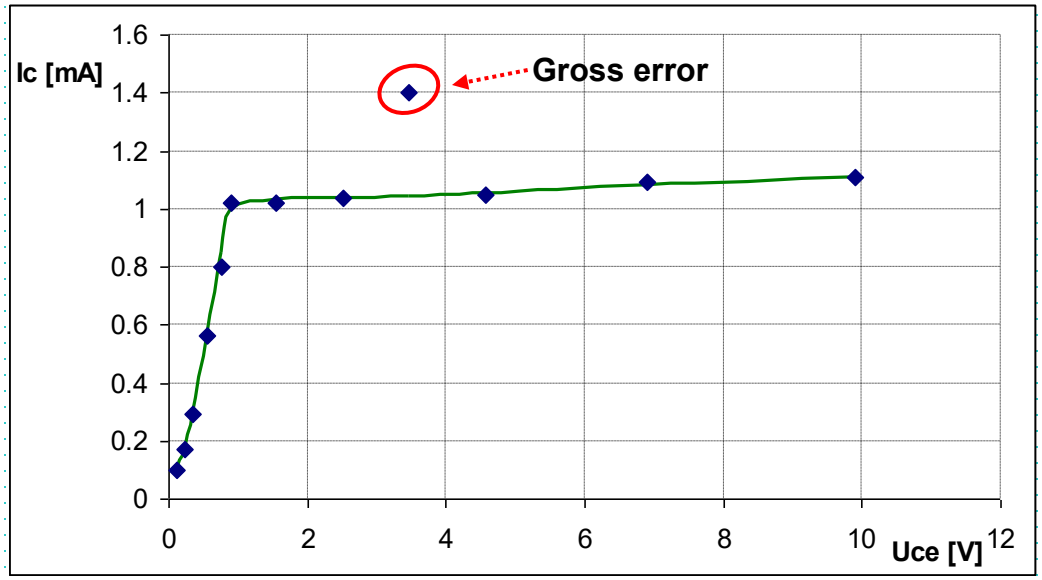


Fig. 1. Output characteristic of npn transistor BC107

Conclusions.

Conclusions can (should) be short (2..4 sentences) but must be specific. In conclusions, evaluate measurements:

- whether performed correctly – if yes, write why you think so, what observed features indicate it e.g., "Results are correct because, consistent with theoretical relation, absolute uncertainty increases with the measured value",

- or not – list errors and try to find their cause. E.g.: comment on omitted points when plotting the characteristic.

- you can also include specific observations and conclusions from obtained results, e.g.: „Measurement on poorly chosen range is burdened with greater uncertainty, which results from comparing plot 1 and 2."

Do not write generalities e.g.: about "loose cables", "non-contacting connections", "inaccurate meters", or about "results consistent with expectations" without justification.